

Technicalities and Results of the Modified RBF Proof of Concept

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Jacobian Determinant Failure under the Modified Map

Why the Jacobian determinant matters for KL-based training. Normalizing flows construct a data-space density by transforming a **base** random variable. Let $Y \in \mathbb{R}^d$ be the base variable with known density p_Y (e.g. $Y \sim \mathcal{N}(0, I)$), and let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a smooth, locally invertible map. The **data** random variable is defined by

$$X = T(Y).$$

To evaluate the model density at a data point x , one maps back to base space via $y = T^{-1}(x)$ and applies change of variables:

$$p_X(x) = p_Y(T^{-1}(x)) |\det \nabla T^{-1}(x)|.$$

Equivalently, taking logs,

$$\log p_X(x) = \log p_Y(T^{-1}(x)) + \log |\det \nabla T^{-1}(x)|.$$

Using $\nabla T^{-1}(x) = (\nabla T(y))^{-1}$ with $y = T^{-1}(x)$ gives the common “forward-Jacobian” form

$$\log p_X(x) = \log p_Y(y) - \log |\det \nabla T(y)|, \quad y = T^{-1}(x).$$

The log-determinant term is the volume-correction factor that makes the transformed density integrate to one. Consequently, training objectives that evaluate $\log p_X$ (including forward-KL estimators implemented through `flow.log_prob`) are numerically well-defined only when $\det J(x)$ stays bounded away from zero on the evaluation set. If $\det J(x) = 0$ at any point, then $\log |\det J(x)| = -\infty$, which drives $\log p_X(x)$ to $-\infty$ and typically results in numerical instability (NaNs) or hard failure.

The modified transform and its Jacobian. Consider the modified map

$$T(z_0) = z_0 + \sum_{k=1}^K \beta_k f(r_k) (z_0 - x_{0k}), \quad r_k := \|z_0 - x_{0k}\|.$$

Each term is a radial displacement from a fixed center x_{0k} with amplitude controlled by a step size β_k and a radial profile $f(\cdot)$. Differentiating with respect to z_0 yields

$$J(z_0) = I + \sum_{k=1}^K \beta_k \left(f(r_k) I + \frac{f'(r_k)}{r_k} (z_0 - x_{0k})(z_0 - x_{0k})^\top \right).$$

This decomposition has a direct geometric interpretation. The $f(r_k)I$ term is an isotropic contribution, scaling derivatives equally in all directions. The second term is rank-one and acts preferentially in the radial direction $(z_0 - x_{0k})$; its sign and magnitude are controlled by $f'(r_k)$ and may be negative when f decreases with radius.

Mechanism for $\det J \leq 0$: eigenvalue crossing induced by the radial term. Local invertibility of T at z_0 is equivalent to $J(z_0)$ being nonsingular. Since

$$\det J(z_0) = \prod_{i=1}^d \lambda_i(z_0),$$

where $\{\lambda_i(z_0)\}$ are the eigenvalues of $J(z_0)$, the event $\det J(z_0) \leq 0$ occurs when at least one eigenvalue reaches 0 (singularity). The Jacobian structure above makes this failure mode natural: even if $f(r_k) > 0$, the rank-one component involving $f'(r_k)$ can drive the derivative in the radial direction downward. When step sizes $\{\beta_k\}$ are sufficiently large (or when multiple terms accumulate coherently), one eigenvalue can be pushed through zero, producing a fold (or near-fold) in the map.

Why $\det J < 0$ is a red flag in this training regime. A negative determinant is not *automatically* forbidden by the change-of-variables formula, because the density uses an absolute value $|\det J|$. If a map were smooth and globally one-to-one, $\det J < 0$ would merely indicate an orientation reversal. The problem here is not the sign by itself, but what the sign *implies about the training path*. During training, the map is not chosen in one shot; it is produced by gradually changing parameters (most importantly the step sizes). At a fixed evaluation point z_0 , the Jacobian determinant varies continuously with these parameters. Therefore, observing a sign flip $\det J(z_0) > 0 \rightarrow \det J(z_0) < 0$ means that $\det J(z_0)$ must have passed through zero at some intermediate parameter value. Passing through $\det J(z_0) = 0$ corresponds to a locally singular Jacobian (an eigenvalue hitting 0), i.e. a local loss of invertibility. Formally, fix z_0 and consider the single-step one-parameter family indexed by a scalar step size β :

$$T_\beta(z_0) = z_0 + \beta f(r)(z_0 - x_0), \quad J_\beta(z_0) = I + \beta A(z_0),$$

where $A(z_0)$ collects the derivative contribution of the update at z_0 . Since $\beta \mapsto J_\beta(z_0)$ is continuous and $\det(\cdot)$ is a polynomial in the matrix entries, the map

$$\beta \mapsto \det J_\beta(z_0)$$

is continuous. Consequently, if $\det J_0(z_0) > 0$ (at $\beta = 0$ the Jacobian is I) and for some β one finds $\det J_\beta(z_0) < 0$, then the intermediate value theorem guarantees the existence of $\beta^* \in (0, \beta)$ such that

$$\det J_{\beta^*}(z_0) = 0.$$

This is why $\det J(z_0) < 0$ is treated as a red flag in practice. In exact arithmetic, if the determinant at a fixed evaluation point z_0 is positive at initialization (where $J = I$ and hence $\det J = 1$) and later becomes negative, then along the continuous training path it must *equal zero at some intermediate parameter value*. That intermediate point corresponds to a singular Jacobian (an eigenvalue hitting 0), i.e. a local loss of invertibility, and it is exactly the situation where the log-determinant contribution becomes ill-defined because $\log |\det J| = \log 0 = -\infty$.

Relation to the Tabak–Turner monotone radial structure. Tabak–Turner’s original construction is designed so that each step behaves like a monotone radial map in the *current* variable, which enables step-size rules that preserve local invertibility more directly. Under the modification where displacements are computed relative to frozen reference locations x_{0k} , that monotone-in-the-current-variable structure is generally lost. As a consequence, local invertibility is no longer an inherited property of the update and must be enforced explicitly; otherwise folds can appear even when pointwise displacements appear well-behaved.

Implemented remedy: backtracking step-size control using slogdet checks. To eliminate the observed determinant sign failures in practice, the fitting procedure was modified to include an explicit *step-size backtracking* stage before accepting each new local map. After computing the nominal Newton step $\beta = -G/H$ (and applying the global cap $|\beta| \leq \epsilon$), the algorithm does *not* immediately apply the update. Instead, it tentatively proposes $\beta_{\text{try}} = \beta$ and verifies that the candidate map preserves local invertibility on a representative subset of points. Concretely, for a small randomly chosen subset $\{z_0^{(i)}\}_{i \in \mathcal{I}}$ (used purely for computational efficiency), the Jacobian of the *current composition plus the candidate map* is assembled and evaluated via a numerically stable determinant routine:

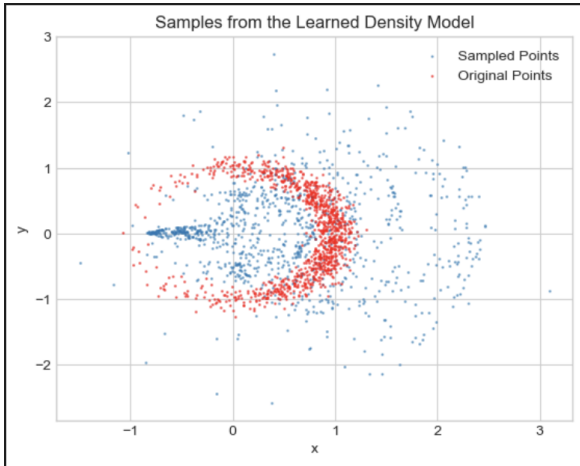
$$(\text{sign}_i, \log |\det J(z_0^{(i)})|) = \text{slogdet}(J(z_0^{(i)})).$$

The acceptance criterion is that the Jacobian be orientation-preserving across the subset, i.e. $\min_{i \in \mathcal{I}} \text{sign}_i > 0$. If this condition fails, the step size is reduced geometrically,

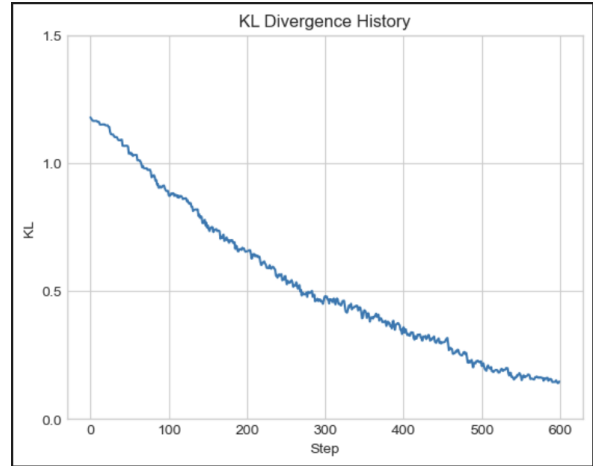
$$\beta_{\text{try}} \leftarrow \frac{1}{2} \beta_{\text{try}},$$

and the Jacobian check is repeated (up to a fixed number of backtracking iterations). If no acceptable β_{try} is found before the step becomes negligible, the candidate map is rejected and the iteration moves on without appending a new transformation. This modification directly targets the failure mechanism described above: it prevents eigenvalues of J from crossing zero (on the checked set) by shrinking the update before folding occurs. In addition, log-density evaluation was made robust by computing log-determinants through slogdet (rather than log det) and explicitly handling the exceptional case $\text{sign} \leq 0$ by returning $\log |\det J| = -\infty$ at those points. While this latter safeguard does not *fix* non-invertibility by itself, it prevents silent numerical corruption (NaNs) and makes determinant violations detectable at the objective level. Together, the backtracking acceptance rule and stable determinant evaluation remove the mid-fit crashes caused by $\log |\det J|$ becoming undefined and enforce the intended diffeomorphic behavior of the learned map on the region used for training.

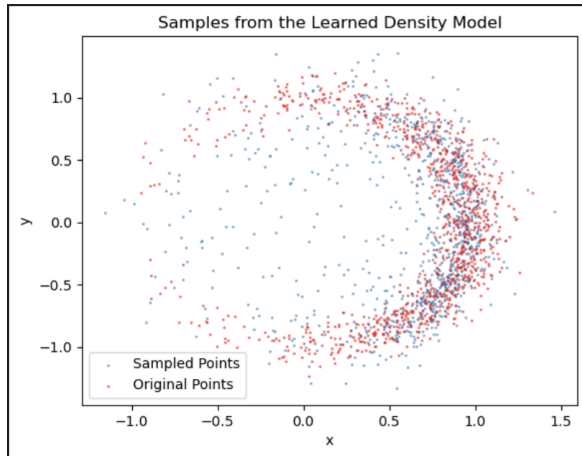
Results



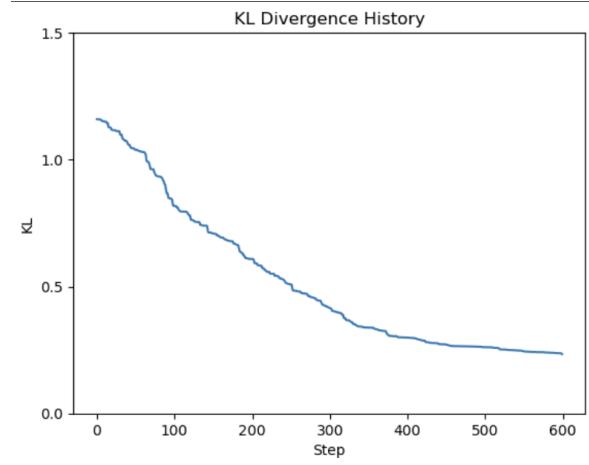
(a) Modified RBF (Test Case 0: `sample_exact(1000)`). Learned samples (blue) overlaid with training data (red).



(b) Modified RBF (Test Case 0: `sample_exact(1000)`). KL divergence history during training.



(a) Tabak-Turner RBF (Test Case 0: `sample_exact(1000)`). Learned samples (blue) overlaid with training data (red).



(b) Tabak-Turner RBF (Test Case 0: `sample_exact(1000)`). KL divergence history during training.